

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 02

PLC Programming Languages

Student Name:	Adamu Nando Usufa
Registration Number:	C11/26/ACIS/17367
Course:	Intro to Critical Infrastructure for IT Learners
Date:	27/09/2026

1. Ladder Diagram

The Ladder Diagram example was opened in OpenPLC Editor. The motor-control logic contains the Start contact, Stop contact, MotorOutput latch and MotorOutput coil.

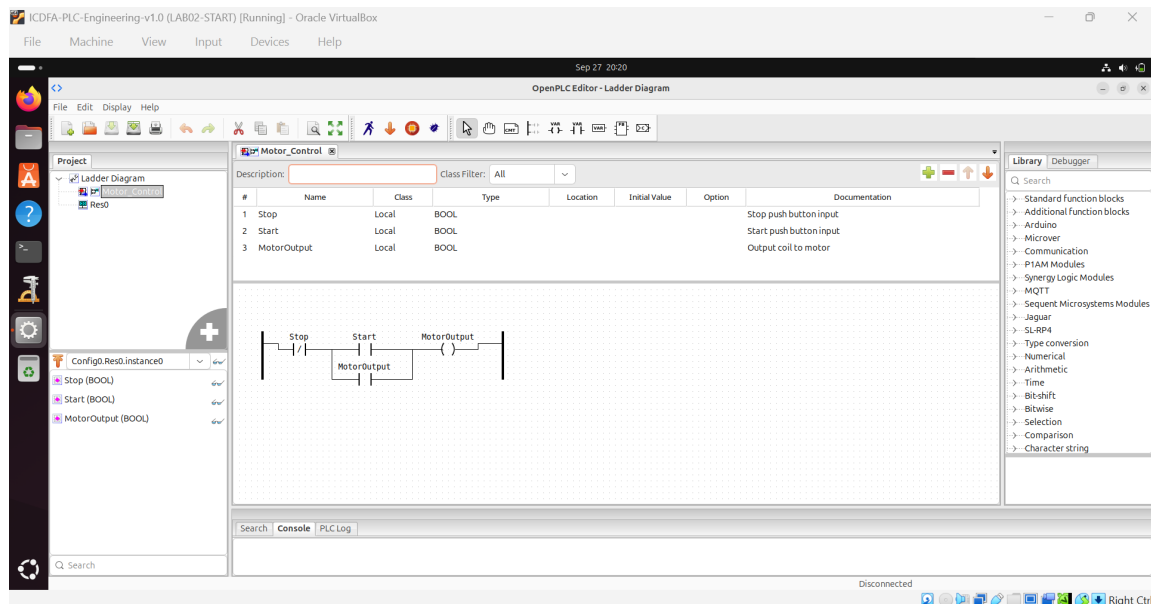


Figure 1: Motor Control Ladder Diagram.

2. Function Block Diagram

The Function Block Diagram represents the same motor start/stop behavior using logical function blocks. The program `Motor_Control_FBD` uses NOT, OR and AND blocks to control `MotorOutput`.

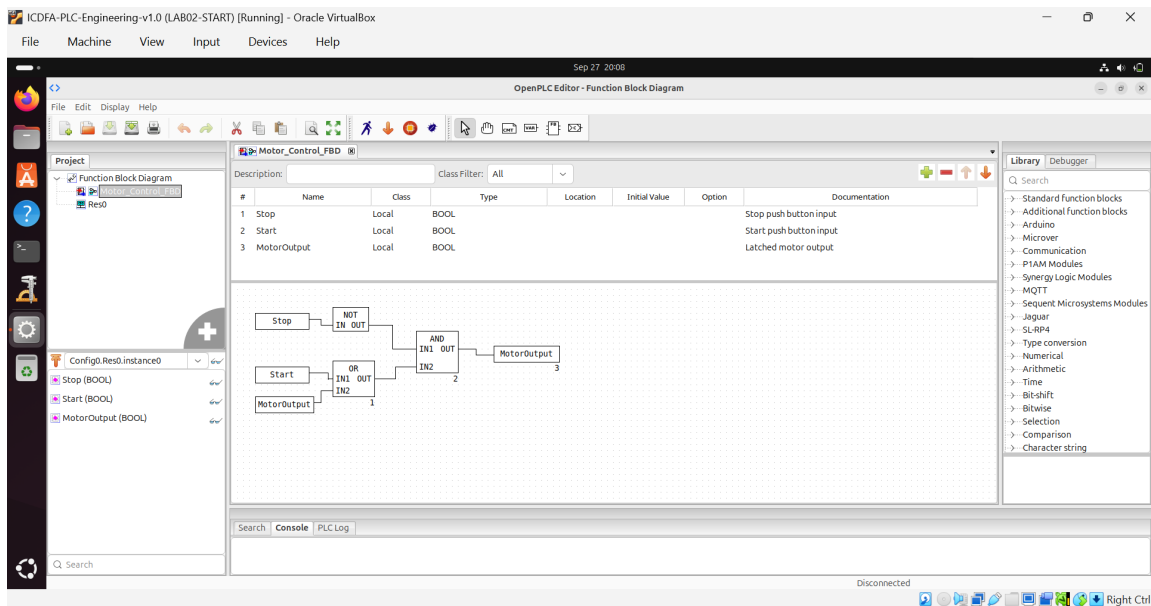


Figure 2: Motor Control Function Block Diagram.

3. Structured Text

The Structured Text example represents the motor-control logic using textual programming. The main expression is:

```
MotorOutput := (NOT Stop) AND (Start OR MotorOutput);
```

The tags Stop, Start and MotorOutput are Boolean values.

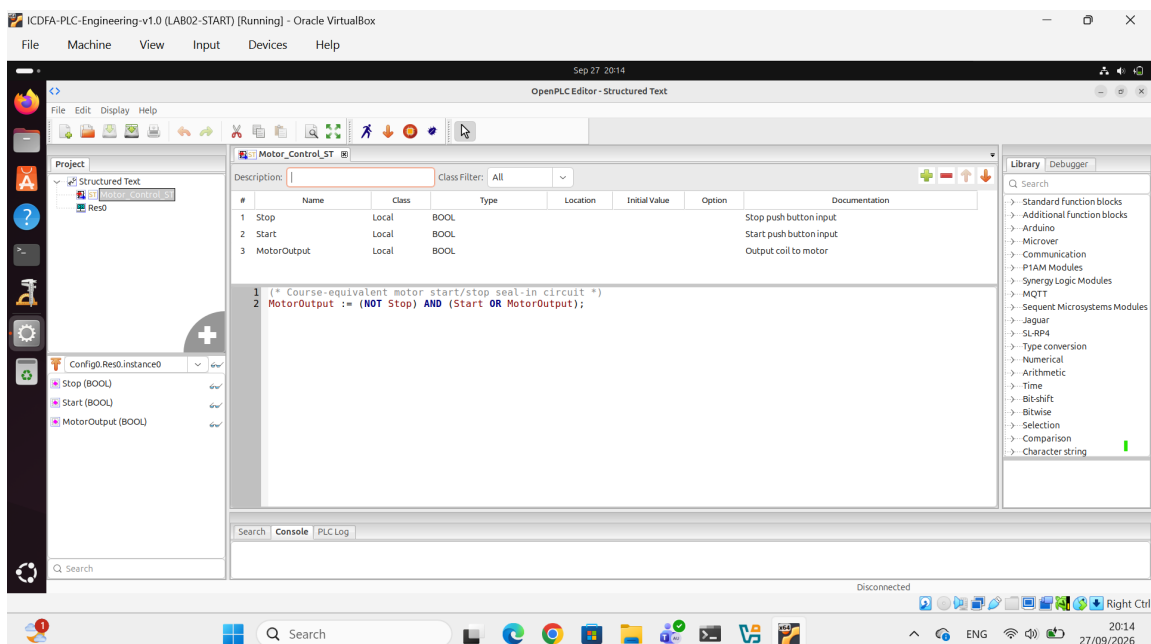


Figure 3: Motor Control Structured Text representation.

4. Instruction List

The Instruction List example represents the control logic using a low-level textual instruction sequence. Compared with Structured Text, Instruction List uses shorter instruction-based statements rather than higher-level expressions.

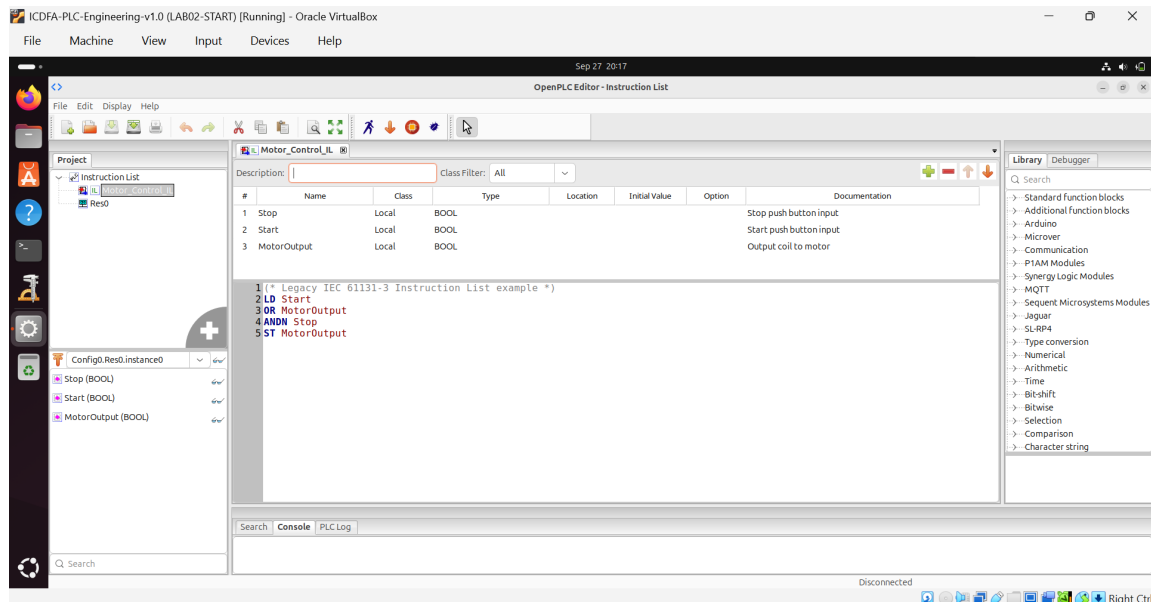


Figure 4: Motor Control Instruction List representation.

5. Sequential Function Chart

The Sequential Function Chart represents the process using steps and transitions. The motor-control example contains an **Idle** state and a **Running** state, with transitions controlled by the Start and Stop conditions.

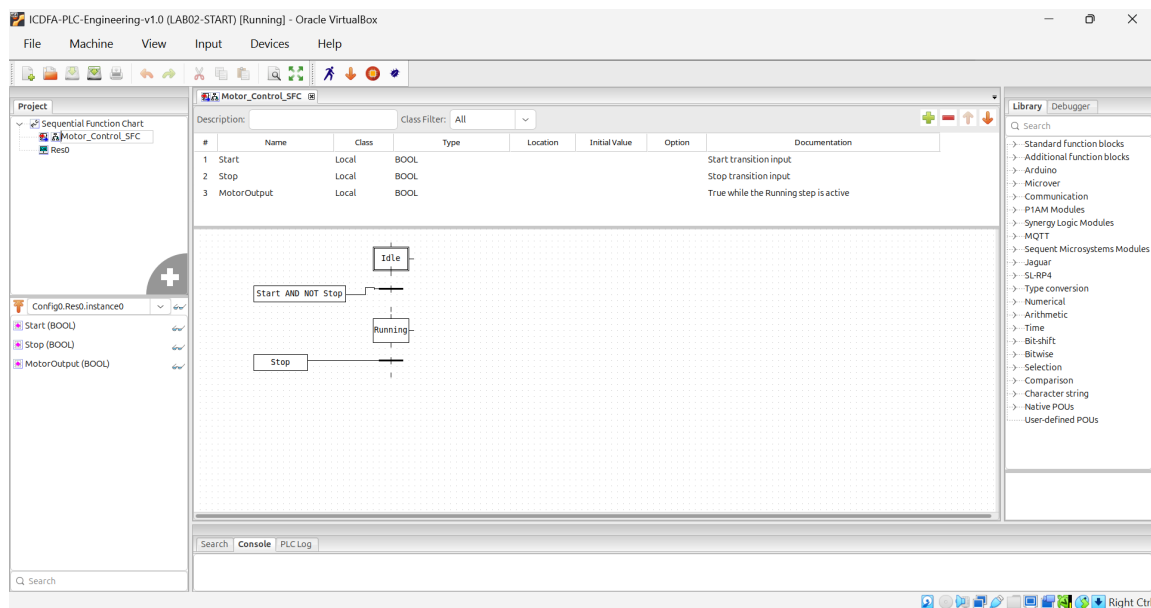


Figure 5: Motor Control Sequential Function Chart.

6. PLC Programming Language Comparison

Representation	Strengths	Limitations	Suitable Industrial Use Case
Ladder Diagram	Easy for technicians familiar with relay logic; visually represents control circuits.	Can become large and difficult to manage for complex logic.	Motor control, interlocks and discrete control systems.
Function Block Diagram	Visual representation of logical and functional relationships.	Large systems may require many interconnected blocks.	Process control and systems using reusable control functions.
Structured Text	Suitable for complex calculations, conditions and text-based logic.	Less intuitive for technicians who prefer graphical logic.	Complex calculations, data processing and advanced control logic.
Instruction List	Compact and low-level representation of PLC instructions.	Less readable and less intuitive for complex programs.	Low-level PLC control and instruction-based programming.
Sequential Function Chart	Clearly represents steps, transitions and process sequences.	More complex to use for simple single-run logic.	Sequential processes, batch operations and machine states.

Table 1: Comparison of PLC programming representations.

7. Equivalent Motor-Control Logic

The same motor start/stop requirement can be represented using different PLC programming languages while maintaining the same control objective.

The basic logic is:

$$\text{MotorOutput} = \text{NOT Stop} \wedge (\text{Start} \vee \text{MotorOutput})$$

In Ladder Diagram, this is represented using contacts and an output coil. In Function Block Diagram, NOT, OR and AND blocks perform the same logic. Structured Text expresses the logic as a Boolean assignment, while Instruction List represents it using sequential instructions. Sequential Function Chart expresses the operation as states and transitions.

Although the representations differ, the intended motor-control behavior remains the same.

8. Knowledge Check

1. Why is Ladder Diagram intuitive for technicians familiar with relay logic?

Ladder Diagram is intuitive because its contacts, coils and control paths resemble traditional electrical relay-control circuits.

2. What is one advantage of Structured Text for complex calculations or conditions?

Structured Text allows complex calculations and conditional logic to be expressed directly using text-based programming statements.

3. When might a Sequential Function Chart be more appropriate than a single ladder rung?

SFC is more appropriate when a process consists of multiple sequential states or steps with defined transitions between them.

4. Why must the meaning of tags remain consistent when the representation/language changes?

Consistent tag meanings ensure that the same inputs, outputs and control variables retain their intended function when the logic is represented in another programming language.

9. Conclusion

The laboratory demonstrated five PLC programming representations: Ladder Diagram, Function Block Diagram, Structured Text, Instruction List and Sequential Function Chart. The same motor start/stop control objective was examined in each representation, showing how PLC logic can be expressed using graphical, textual and sequential approaches.